

26.1 Basic configuration

The QEI is configured using the following registers:

1. Power: In the PCONP register ([Table 46](#)), set bit PCQEI.
Remark: On reset, the QEI is disabled (PCQEI = 0).
2. Peripheral clock: In the PCLKSEL0 register ([Table 40](#)), select PCLK_QEI.
3. Pins: Select QEI pins through the PINSEL registers. Select pin modes for port pins with QEI functions through the PINMODE registers ([Section 8.5](#)).
4. Interrupts: See [Section 26.6.4](#). The QEI interrupt is enabled in the NVIC using the appropriate Interrupt Set Enable register.

26.2 Features

This Quadrature Encoder Interface (QEI) has the following features:

- tracks encoder position.
- increments/ decrements depending on direction.
- programmable for 2X or 4X position counting.
- velocity capture using built-in timer.
- velocity compare function with less than interrupt.
- uses 32-bit registers for position and velocity.
- three position compare registers with interrupts.
- index counter for revolution counting.
- index compare register with interrupts.
- can combine index and position interrupts to produce an interrupt for whole and partial revolution displacement.
- digital filter with programmable delays for encoder input signals.
- can accept decoded signal inputs (clock and direction).

26.3 Introduction

A quadrature encoder, also known as a 2-channel incremental encoder, converts angular displacement into two pulse signals. By monitoring both the number of pulses and the relative phase of the two signals, you can track the position, direction of rotation, and velocity. In addition, a third channel, or index signal, can be used to reset the position counter. This quadrature encoder interface module decodes the digital pulses from a quadrature encoder wheel to integrate position over time and determine direction of rotation. In addition, it can capture the velocity of the encoder wheel.

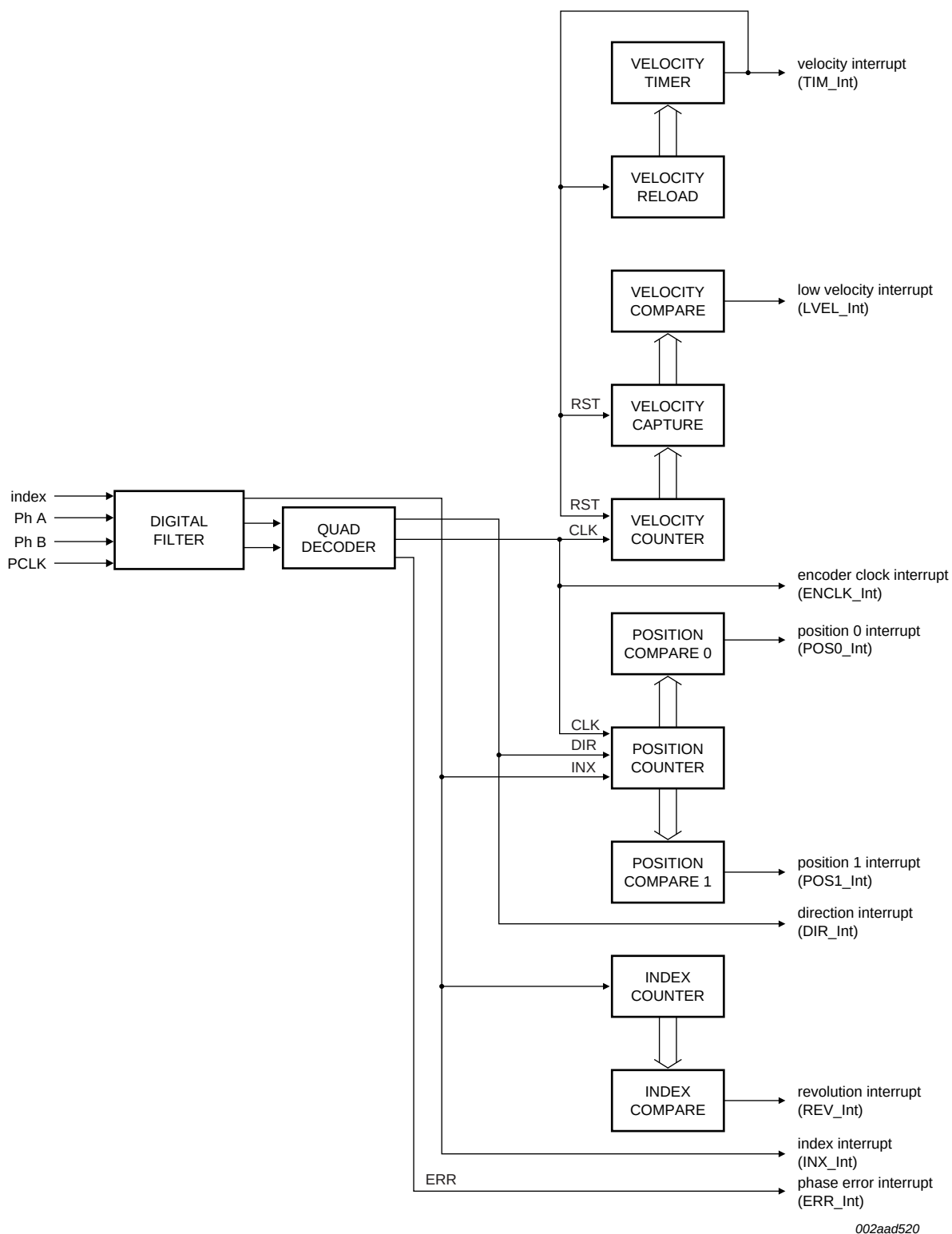


Fig 129.Encoder interface block diagram

26.4 Functional description

The QEI module interprets the two-bit gray code produced by a quadrature encoder wheel to integrate position over time and determine direction of rotation. In addition, it can capture the velocity of the encoder wheel.

26.4.1 Input signals

The QEI module supports two modes of signal operation: quadrature phase mode and clock/direction mode. In quadrature phase mode, the encoder produces two clocks that are 90 degrees out of phase; the edge relationship is used to determine the direction of rotation. In clock/direction mode, the encoder produces a clock signal to indicate steps and a direction signal to indicate the direction of rotation.).

This mode is determined by the SigMode bit of the QEI Configuration (QEICONF) register (See [Table 486](#)). When the SigMode bit = 1, the quadrature decoder is bypassed and the PhA pin functions as the direction signal and PhB pin functions as the clock signal for the counters, etc. When the SigMode bit = 0, the PhA pin and PhB pins are decoded by the quadrature decoder. In this mode the quadrature decoder produces the direction and clock signals for the counters, etc. In both modes the direction signal is subject to the effects of the direction invert (DIRINV) bit.

26.4.1.1 Quadrature input signals

When edges on PhA lead edges on PhB, the position counter is incremented. When edges on PhB lead edges on PhA, the position counter is decremented. When a rising and falling edge pair is seen on one of the phases without any edges on the other, the direction of rotation has changed.

Table 480. Encoder states

Phase A	Phase B	state
1	0	1
1	1	2
0	1	3
0	0	4

Table 481. Encoder state transitions^[1]

from state	to state	Direction
1	2	positive
2	3	
3	4	
4	1	negative
4	3	
3	2	
2	1	
1	4	

[1] All other state transitions are illegal and should set the ERR bit.

Interchanging of the PhA and PhB input signals are compensated by complementing the DIR bit. When set = 1, the direction inversion bit (DIRINV) complements the DIR bit.

Table 482. Encoder direction

DIR bit	DIRINV bit	direction
0	0	forward
1	0	reverse
0	1	reverse
1	1	forward

Figure 130 shows how quadrature encoder signals equate to direction and count.

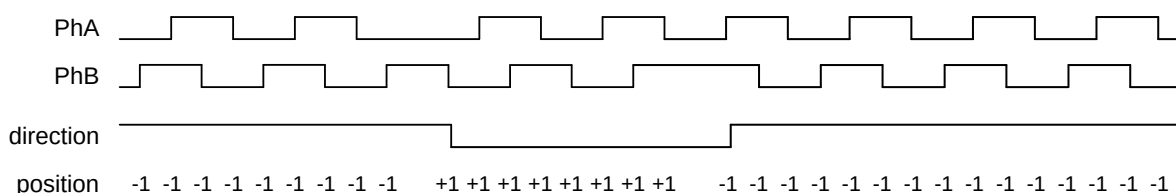


Fig 130. Quadrature Encoder Basic Operation

26.4.1.2 Digital input filtering

All three encoder inputs (PhA, PhB, and index) require digital filtering. The number of sample clocks is user programmable from 1 to 4,294,967,295 (0xFFFF FFFF). In order for a transition to be accepted, the input signal must remain in new state for the programmed number of sample clocks.

26.4.2 Position capture

The capture mode for the position integrator can be set to update the position counter on every edge of the PhA signal or to update on every edge of both PhA and PhB. Updating the position counter on every PhA and PhB provides more positional resolution at the cost of less range in the positional counter.

The position integrator and velocity capture can be independently enabled. Alternatively, the phase signals can be interpreted as a clock and direction signal as output by some encoders.

The position counter is automatically reset on one of two conditions. Incrementing past the maximum position value (QEIMAXPOS) will reset the position counter to zero. If the reset on index bit (RESPI) is set, sensing the index pulse will reset the position counter to zero.

26.4.3 Velocity capture

The velocity capture has a programmable timer and a capture register. It counts the number of phase edges (using the same configuration as for the position integrator) in a given time period. When the velocity timer (QEITIME) overflows the contents of the velocity counter (QEIVEL) are transferred to the capture (QEICAP) register. The velocity counter is then cleared. The velocity timer is loaded with the contents of the velocity reload register (QEILOAD). Finally, the velocity interrupt (TIM_Int) is asserted. The

number of edges counted in a given time period is directly proportional to the velocity of the encoder. Setting the reset velocity bit (RESV) has the same effect as an overflow of the velocity timer, except that the setting the RESV bit will not generate a velocity interrupt.

The following equation converts the velocity counter value into an RPM value:

$$\text{RPM} = (\text{PCLK} * \text{QEICAP} * 60) \div (\text{QEILOAD} * \text{PPR} * \text{Edges})$$

where:

- **PCLK** is the peripheral clock rate for the QEI block. See [Section 4.7.3](#) for more on the possibilities for PCLK).
- **QEICAP** is the captured velocity counter value for the last velocity timer period.
- **QEILOAD** is the velocity timer reload value.
- **PPR** is the number of pulses per revolution of the physical encoder used in the application
- **Edges** is 2 or 4, based on the capture mode set in the QEICON register (2 for CapMode set to 0 and 4 for CapMode set to 1)

For example, consider a motor running at 600 RPM. A 2048 pulse per revolution quadrature encoder is attached to the motor, producing 8192 phase edges per revolution (PPR * Edges). This results in 81,920 pulses per second (the motor turns 10 times per second at 600 RPM and there are 8,192 edges per revolution). If the timer were clocked at 10,000 Hz, and the QEILOAD was 2,500 (corresponding to ¼ of a second), it would count 20,480 pulses per update. Using the above equation:

$$\text{RPM} = (10000 * 1 * 20480 * 60) \div (2500 * 2048 * 4) = 600 \text{ RPM}$$

Now, consider that the motor is sped up to 3000 RPM. This results in 409,600 pulses per second, or 102,400 every ¼ of a second. Again, the above equation gives:

$$\text{RPM} = (10000 * 1 * 102400 * 60) \div (2500 * 2048 * 4) = 3000 \text{ RPM}$$

These are simple examples, real-world values will have a higher rate for PCLK, and probably a larger value for QEILOAD as well.

26.4.4 Velocity compare

In addition to velocity capture, the velocity measurement system includes a programmable velocity compare register. After every velocity capture event the contents of the velocity capture register (QEICAP) is compared with the contents of the velocity compare register (VELCOMP). If the captured velocity is less than the compare value an interrupt is asserted provided that the velocity compare interrupt enable bit is set. This can be used to determine if a motor shaft is either stalled or moving too slow.

26.5 Pin description

Table 483. QEI pin description

Pin name	I/O	Description
MCI0 [1]	I	Used as the Phase A (PhA) input to the Quadrature Encoder Interface.
MCI1 [1]	I	Used as the Phase B (PhB) input to the Quadrature Encoder Interface.
MCI2 [1]	I	Used as the Index (IDX) input to the Quadrature Encoder Interface.

- [1] The Quadrature Encoder Interface uses the same pin functions as the Motor Control PWM feedback inputs and are connected when the Motor Control PWM function is selected on these pins. If used as part of motor control, the QEI is an alternative to feedback directly to the MCPWM.

26.6 Register description

26.6.1 Register summary

Table 484. QEI Register summary

Name	Description	Access	Reset value	Address
Control registers				
QEICON	Control register	WO	0	0x400B C000
QEICONF	Configuration register	R/W	0	0x400B C008
QEISTAT	Encoder status register	RO	0	0x400B C004
Position, index, and timer registers				
QEIPOS	Position register	RO	0	0x400B C00C
QEIMAXPOS	Maximum position register	R/W	0	0x400B C010
CMPOS0	position compare register 0	R/W	0	0x400B C014
CMPOS1	position compare register 1	R/W	0	0x400B C018
CMPOS2	position compare register 2	R/W	0	0x400B C01C
INXCNT	Index count register	RO	0	0x400B C020
INXCMP	Index compare register	R/W	0	0x400B C024
QEILOAD	Velocity timer reload register	R/W	0	0x400B C028
QEITIME	Velocity timer register	RO	0	0x400B C02C
QEIVEL	Velocity counter register	RO	0	0x400B C030
QEICAP	Velocity capture register	RO	0	0x400B C034
VELCOMP	Velocity compare register	R/W	0	0x400B C038
FILTER	Digital filter register	R/W	0	0x400B C03C
Interrupt registers				
QEINTSTAT	Interrupt status register	RO	0	0x400B CFE0
QEISET	Interrupt status set register	WO	0	0x400B CFEC
QEICLR	Interrupt status clear register	WO	0	0x400B CFE8
QEIIIE	Interrupt enable register	RO	0	0x400B CFE4
QEIIES	Interrupt enable set register	WO	0	0x400B CFDC
QEIIEC	Interrupt enable clear register	WO	0	0x400B CFD8

26.6.2 Control registers

26.6.2.1 QEI Control register (QEICON - 0x400B C000)

This register contains bits which control the operation of the position and velocity counters of the QEI module.

Table 485: QEI Control register (QEICON - address 0x400B C000) bit description

Bit	Symbol	Description	Reset value
0	RESP	Reset position counter. When set = 1, resets the position counter to all zeros. Autoclears when the position counter is cleared.	0
1	RESPI	Reset position counter on index. When set = 1, resets the position counter to all zeros when an index pulse occurs. Autoclears when the position counter is cleared.	0
2	RESV	Reset velocity. When set = 1, resets the velocity counter to all zeros and reloads the velocity timer. Autoclears when the velocity counter is cleared.	0
3	RESI	Reset index counter. When set = 1, resets the index counter to all zeros. Autoclears when the index counter is cleared.	0
31:4	-	reserved	0

26.6.2.2 QEI Configuration register (QEICONF - 0x400B C008)

This register contains the configuration of the QEI module.

Table 486: QEI Configuration register (QEICONF - address 0x400B C008) bit description

Bit	Symbol	Description	Reset value
0	DIRINV	Direction invert. When = 1, complements the DIR bit.	0
1	SIGMODE	Signal Mode. When = 0, PhA and PhB function as quadrature encoder inputs. When = 1, PhA functions as the direction signal and PhB functions as the clock signal.	0
2	CAPMODE	Capture Mode. When = 0, only PhA edges are counted (2X). When = 1, BOTH PhA and PhB edges are counted (4X), increasing resolution but decreasing range.	0
3	INVINX	Invert Index. When set, inverts the sense of the index input.	0
31:4	-	reserved	0

26.6.2.3 QEI Status register (QEISTAT - 0x400B C004)

This register provides the status of the encoder interface.

Table 487: QEI Interrupt Status register (QEISTAT - address 0x400B C004) bit description

Bit	Symbol	Description	Reset value
0	DIR	Direction bit. In combination with DIRINV bit indicates forward or reverse direction. See Table 482 .	0
31:1	-	reserved	0

26.6.3 Position, index and timer registers

26.6.3.1 QEI Position register (QEIPPOS - 0x400B C00C)

This register contains the current value of the encoder position. Increments or decrements when encoder counts occur, depending on the direction of rotation.

Table 488: QEI Position register (QEIPPOS - address 0x400B C00C) bit description

Bit	Symbol	Description	Reset value
31:0	-	Current position value.	0

26.6.3.2 QEI Maximum Position register (QEIMAXPOS - 0x400B C010)

This register contains the maximum value of the encoder position. In forward rotation the position register resets to zero when the position register exceeds this value. In reverse rotation the position register resets to this value when the position register decrements from zero.

Table 489: QEI Maximum Position register (QEIMAXPOS - address 0x400B C010) bit description

Bit	Symbol	Description	Reset value
31:0	-	Current maximum position value.	0

26.6.3.3 QEI Position Compare register 0 (CMPOS0 - 0x400B C014)

This register contains a position compare value. This value is compared against the current value of the position register. Interrupts can be enabled to interrupt when the compare value is equal to the current value of the position register.

Table 490: QEI Position Compare register 0 (CMPOS0 - address 0x400B C014) bit description

Bit	Symbol	Description	Reset value
31:0	-	Position compare value 0.	0

26.6.3.4 QEI Position Compare register 1 (CMPOS1 - 0x400B C018)

This register contains a position compare value. This value is compared against the current value of the position register. Interrupts can be enabled to interrupt when the compare value is equal to the current value of the position register.

Table 491: QEI Position Compare register 1 (CMPOS1 - address 0x400B C018) bit description

Bit	Symbol	Description	Reset value
31:0	-	Position compare value 1.	0

26.6.3.5 QEI Position Compare register 2 (CMPOS2 - 0x400B C01C)

This register contains a position compare value. This value is compared against the current value of the position register. Interrupts can be enabled to interrupt when the compare value is equal to the current value of the position register.

Table 492: QEI Position Compare register 2 (CMPOS2 - address 0x400B C01C) bit description

Bit	Symbol	Description	Reset value
31:0	-	Position compare value 2.	0

26.6.3.6 QEI Index Count register (INXCNT - 0x400B C020)

This register contains the current value of the index counter. It is updated when an index count occurs. This can be an increment when the position counter overflows the MAXPOS value or a decrement when the position counter underflows zero, depending on the direction of rotation. In case (re)calibration occurs due to an index pulse, the over/underflow is forced internally.

Table 493: QEI Index Count register (INXCNT - address 0x400B C020) bit description

Bit	Symbol	Description	Reset value
31:0	-	Current index counter value.	0

26.6.3.7 QEI Index Compare register (INXCMP - 0x400B C024)

This register contains an index compare value. This value is compared against the current value of the index count register. Interrupts can be enabled to interrupt when the compare value is equal to the current value of the index count register.

Table 494: QEI Index Compare register (INXCMP - address 0x400B C024) bit description

Bit	Symbol	Description	Reset value
31:0	-	Index compare value.	0

26.6.3.8 QEI Timer Reload register (QEILOAD - 0x400B C028)

This register contains the reload value of the velocity timer. When the timer (QEITIME) overflows or the RESV bit is asserted, this value is loaded into the timer (QEITIME).

Table 495: QEI Timer Load register (QEILOAD - address 0x400B C028) bit description

Bit	Symbol	Description	Reset value
31:0	-	Velocity timer reload value.	0

26.6.3.9 QEI Timer register (QEITIME - 0x400B C02C)

This register contains the current value of the velocity timer. When this timer overflows the value of velocity counter (QEIVEL) is stored in the velocity capture register (QEICAP), the velocity counter is reset to zero, the timer is reloaded with the value stored in the velocity reload register (QEILOAD), and the velocity interrupt (TIM_Int) is asserted.

Table 496: QEI Timer register (QEITIME - address 0x400B C02C) bit description

Bit	Symbol	Description	Reset value
31:0	-	Current velocity timer value.	0

26.6.3.10 QEI Velocity register (QEIVEL - 0x400B C030)

This register contains the running count of velocity pulses for the current time period. When the velocity timer (QEITIME) overflows the contents of this register is captured in the velocity capture register (QEICAP). After capture, this register is set to zero. This register is also reset when the velocity reset bit (RESV) is asserted.

Table 497: QEI Velocity register (QEIVEL - address 0x400B C030) bit description

Bit	Symbol	Description	Reset value
31:0	-	Current velocity pulse count.	0

26.6.3.11 QEI Velocity Capture register (QEICAP - 0x400B C034)

This register contains the most recently measured velocity of the encoder. This corresponds to the number of velocity pulses counted in the previous velocity timer period. The current velocity count is latched into this register when the velocity timer overflows.

Table 498: QEI Velocity Capture register (QEICAP - address 0x400B C034) bit description

Bit	Symbol	Description	Reset value
31:0	-	Captured velocity pulse count.	0

26.6.3.12 QEI Velocity Compare register (VELCOMP - 0x400B C038)

This register contains a velocity compare value. This value is compared against the captured velocity in the velocity capture register. If the capture velocity is less than the value in this compare register, a velocity compare interrupt (VELC_Int) will be asserted, if enabled.

Table 499: QEI Velocity Compare register (VELCOMP - address 0x400B C038) bit description

Bit	Symbol	Description	Reset value
31:0	-	Velocity compare value.	0

26.6.3.13 QEI Digital Filter register (FILTER - 0x400B C03C)

This register contains the sampling count for the digital filter. A sampling count of zero bypasses the filter.

Table 500: QEI Digital Filter register (FILTER - address 0x400B C03C) bit description

Bit	Symbol	Description	Reset value
31:0	-	Digital filter sampling delay	0x0

26.6.4 Interrupt registers

26.6.4.1 QEI Interrupt Status register (QEINTSTAT)

This register provides the status of the encoder interface and the current set of interrupt sources that are asserted to the controller. Bits set to 1 indicate the latched events that have occurred; a zero bit indicates that the event in question has not occurred. Writing a 0 to a bit position clears the corresponding interrupt.

Table 501: QEI Interrupt Status register (QEINTSTAT - address 0x400B CFE0) bit description

Bit	Symbol	Description	Reset value
0	INX_Int	Indicates that an index pulse was detected.	0
1	TIM_Int	Indicates that a velocity timer overflow occurred	0
2	VELC_Int	Indicates that captured velocity is less than compare velocity.	0
3	DIR_Int	Indicates that a change of direction was detected.	0
4	ERR_Int	Indicates that an encoder phase error was detected.	0
5	ENCLK_Int	Indicates that an encoder clock pulse was detected.	
6	POS0_Int	Indicates that the position 0 compare value is equal to the current position.	0
7	POS1_Int	Indicates that the position 1 compare value is equal to the current position.	0
8	POS2_Int	Indicates that the position 2 compare value is equal to the current position.	0
9	REV_Int	Indicates that the index compare value is equal to the current index count.	0
10	POS0REV_Int	Combined position 0 and revolution count interrupt. Set when both the POS0_Int bit is set and the REV_Int is set.	0
11	POS1REV_Int	Combined position 1 and revolution count interrupt. Set when both the POS1_Int bit is set and the REV_Int is set.	0
12	POS2REV_Int	Combined position 2 and revolution count interrupt. Set when both the POS2_Int bit is set and the REV_Int is set.	0
31:13	-	reserved	0

26.6.4.2 QEI Interrupt Set register (QEISSET - 0x400B CFEC)

Writing a one to a bit in this register sets the corresponding bit in the QEI Interrupt Status register (QEISTAT).

Table 502: QEI Interrupt Set register (QEISSET - address 0x400B CFEC) bit description

Bit	Symbol	Description	Reset value
0	INX_Int	Indicates that an index pulse was detected.	0
1	TIM_Int	Indicates that a velocity timer overflow occurred	0
2	VELC_Int	Indicates that captured velocity is less than compare velocity.	0
3	DIR_Int	Indicates that a change of direction was detected.	0
4	ERR_Int	Indicates that an encoder phase error was detected.	0
5	ENCLK_Int	Indicates that an encoder clock pulse was detected.	
6	POS0_Int	Indicates that the position 0 compare value is equal to the current position.	0
7	POS1_Int	Indicates that the position 1 compare value is equal to the current position.	0
8	POS2_Int	Indicates that the position 2 compare value is equal to the current position.	0
9	REV_Int	Indicates that the index compare value is equal to the current index count.	0

Table 502: QEI Interrupt Set register (QEISSET - address 0x400B CFEC) bit description ...continued

Bit	Symbol	Description	Reset value
10	POS0REV_Int	Combined position 0 and revolution count interrupt. Set when both the POS0_Int bit is set and the REV_Int is set.	0
11	POS1REV_Int	Combined position 1 and revolution count interrupt. Set when both the POS1_Int bit is set and the REV_Int is set.	0
12	POS2REV_Int	Combined position 2 and revolution count interrupt. Set when both the POS2_Int bit is set and the REV_Int is set.	0
31:13	-	reserved	0

26.6.4.3 QEI Interrupt Clear register (QEICLR - 0x400B CFE8)

Writing a 1 to a bit in this register clears the corresponding bit in the QEI Interrupt Status register (QEISTAT).

Table 503: QEI Interrupt Clear register (QEICLR - 0x400B CFE8) bit description

Bit	Symbol	Description	Reset value
0	INX_Int	Indicates that an index pulse was detected.	0
1	TIM_Int	Indicates that a velocity timer overflow occurred	0
2	VELC_Int	Indicates that captured velocity is less than compare velocity.	0
3	DIR_Int	Indicates that a change of direction was detected.	0
4	ERR_Int	Indicates that an encoder phase error was detected.	0
5	ENCLK_Int	Indicates that an encoder clock pulse was detected.	0
6	POS0_Int	Indicates that the position 0 compare value is equal to the current position.	0
7	POS1_Int	Indicates that the position 1 compare value is equal to the current position.	0
8	POS2_Int	Indicates that the position 2 compare value is equal to the current position.	0
9	REV_Int	Indicates that the index compare value is equal to the current index count.	0
10	POS0REV_Int	Combined position 0 and revolution count interrupt. Set when both the POS0_Int bit is set and the REV_Int is set.	0
11	POS1REV_Int	Combined position 1 and revolution count interrupt. Set when both the POS1_Int bit is set and the REV_Int is set.	0
12	POS2REV_Int	Combined position 2 and revolution count interrupt. Set when both the POS2_Int bit is set and the REV_Int is set.	0
31:13	-	reserved	0

26.6.4.4 QEI Interrupt Enable register (QEIE - 0x400B CFE4)

This register enables interrupt sources. Bits set to 1 enable the corresponding interrupt; a 0 bit disables the corresponding interrupt.

Table 504: QEI Interrupt Enable register (QEIE - address 0x400B CFE4) bit description

Bit	Symbol	Description	Reset value
0	INX_Int	Indicates that an index pulse was detected.	0
1	TIM_Int	Indicates that a velocity timer overflow occurred	0
2	VELC_Int	Indicates that captured velocity is less than compare velocity.	0
3	DIR_Int	Indicates that a change of direction was detected.	0

Table 504: QEI Interrupt Enable register (QEIE - address 0x400B CFE4) bit description ...continued

Bit	Symbol	Description	Reset value
4	ERR_Int	Indicates that an encoder phase error was detected.	0
5	ENCLK_Int	Indicates that an encoder clock pulse was detected.	0
6	POS0_Int	Indicates that the position 0 compare value is equal to the current position.	0
7	POS1_Int	Indicates that the position 1 compare value is equal to the current position.	0
8	POS2_Int	Indicates that the position 2 compare value is equal to the current position.	0
9	REV_Int	Indicates that the index compare value is equal to the current index count.	0
10	POS0REV_Int	Combined position 0 and revolution count interrupt. Set when both the POS0_Int bit is set and the REV_Int is set.	0
11	POS1REV_Int	Combined position 1 and revolution count interrupt. Set when both the POS1_Int bit is set and the REV_Int is set.	0
12	POS2REV_Int	Combined position 2 and revolution count interrupt. Set when both the POS2_Int bit is set and the REV_Int is set.	0
31:13	-	reserved	0

26.6.4.5 QEI Interrupt Enable Set register (QEIES - 0x400B CFDC)

Writing a 1 to a bit in this register sets the corresponding bit in the QEI Interrupt Enable register (QEIE).

Table 505: QEI Interrupt Enable Set register (QEIES - address 0x400B CFDC) bit description

Bit	Symbol	Description	Reset value
0	INX_EN	Indicates that an index pulse was detected.	0
1	TIM_EN	Indicates that a velocity timer overflow occurred	0
2	VELC_EN	Indicates that captured velocity is less than compare velocity.	0
3	DIR_EN	Indicates that a change of direction was detected.	0
4	ERR_EN	Indicates that an encoder phase error was detected.	0
5	ENCLK_EN	Indicates that an encoder clock pulse was detected.	0
6	POS0_Int	Indicates that the position 0 compare value is equal to the current position.	0
7	POS1_Int	Indicates that the position 1 compare value is equal to the current position.	0
8	POS2_Int	Indicates that the position 2 compare value is equal to the current position.	0
9	REV_Int	Indicates that the index compare value is equal to the current index count.	0
10	POS0REV_Int	Combined position 0 and revolution count interrupt. Set when both the POS0_Int bit is set and the REV_Int is set.	0
11	POS1REV_Int	Combined position 1 and revolution count interrupt. Set when both the POS1_Int bit is set and the REV_Int is set.	0
12	POS2REV_Int	Combined position 2 and revolution count interrupt. Set when both the POS2_Int bit is set and the REV_Int is set.	0
31:13	-	reserved	0

26.6.4.6 QEI Interrupt Enable Clear register (QEIEC - 0x400B CFD8)

Writing a 1 to a bit in this register clears the corresponding bit in the QEI Interrupt Enable register (QEIE).

Table 506: QEI Interrupt Enable Clear register (QEIEC - address 0x400B CFD8) bit description

Bit	Symbol	Description	Reset value
0	INX_EN	Indicates that an index pulse was detected.	0
1	TIM_EN	Indicates that a velocity timer overflow occurred	0
2	VELC_EN	Indicates that captured velocity is less than compare velocity.	0
3	DIR_EN	Indicates that a change of direction was detected.	0
4	ERR_EN	Indicates that an encoder phase error was detected.	0
5	ENCLK_EN	Indicates that an encoder clock pulse was detected.	0
6	POS0_Int	Indicates that the position 0 compare value is equal to the current position.	0
7	POS1_Int	Indicates that the position 1 compare value is equal to the current position.	0
8	POS2_Int	Indicates that the position 2 compare value is equal to the current position.	0
9	REV_Int	Indicates that the index compare value is equal to the current index count.	0
10	POS0REV_Int	Combined position 0 and revolution count interrupt. Set when both the POS0_Int bit is set and the REV_Int is set.	0
11	POS1REV_Int	Combined position 1 and revolution count interrupt. Set when both the POS1_Int bit is set and the REV_Int is set.	0
12	POS2REV_Int	Combined position 2 and revolution count interrupt. Set when both the POS2_Int bit is set and the REV_Int is set.	0
31:13	-	reserved	0